Ebbeju Lankapalli

Aspiring Artificial Intelligence & Machine Learning Engineer

Vijayawada, Andhra Pradesh, India ·

ebbejulankapalli@gmail.com · +91 7396774148 · GitHub ·

LinkedIn

SUMMARY

- B.Tech CSE (AI & ML) student with hands-on experience building end-to-end Machine Learning, Generative AI, and Agentic AI applications. Experienced in supervised and unsupervised learning, deep learning, LLMs, RAG, AI agents, MCP, model evaluation, and full-stack AI development. Proficient in Python, FastAPI, React, LangChain, LangGraph, Google Gemini, CatBoost, PostgreSQL, MongoDB, Docker, and modern AI tooling, with experience developing and deploying production-oriented AI systems.

EDUCATION

- **B.Tech in Computer Science and Engineering (AI & ML)** **(2023 - 2027)**
  Dhanekula Institute of Engineering and Technology, Vijayawada, AP
  CGPA: 8.2/10

SKILLS

- **Programming Languages**
  Python, C, C++, SQL, JavaScript, TypeScript
- **Machine Learning & Deep Learning**
  Supervised Learning, Unsupervised Learning, Regression, Classification, Ensemble Learning, Clustering, Feature Engineering, Model Evaluation, Scikit-learn, CatBoost, PyTorch, NumPy, Pandas, Matplotlib
- **Generative AI & Agentic AI**
  LLMs, Generative AI, RAG, Semantic Search, Vector Embeddings, AI Agents, LangChain, LangGraph, MCP, Google Gemini, Groq, QLoRA, LoRA, PEFT, Prompt Engineering
- **Backend, Full-Stack & Databases**
  FastAPI, REST APIs, React, Vite, Node.js, Express.js, PostgreSQL, MongoDB, Supabase, SQLAlchemy
- **DevOps, Testing & Developer Tools**
  Docker, Redis, Celery, Git, GitHub, GitHub Actions, Pytest, Ruff, VS Code, Jupyter Notebook, Google Colab, Anaconda

PROJECTS

- **Financial Fraud Detection System — End-to-End AI/ML Application** GitHub · Live Demo
  - Built and deployed an end-to-end fraud detection platform using CatBoost on the IEEE-CIS Fraud Detection dataset, achieving 96.91% accuracy, 58.02% precision, 40.61% recall, 47.78% F1, and 88.51% ROC-AUC on a chronological held-out test set.
  - Developed a 20-feature production inference pipeline with validation-based threshold optimization (0.83), FastAPI APIs, PostgreSQL transaction persistence, React analytics dashboard, Dockerized services, and Vercel/Render deployment.
  - Integrated a Groq-powered GPT-OSS-20B Fraud Investigation Agent for grounded explanations and investigation guidance while keeping CatBoost as the authoritative fraud decision layer; added Qwen2.5 + QLoRA research experiments, grounding safeguards, and 21 passing backend tests.

- **OmniLead AI — AI-Powered Omnichannel Lead Intelligence Platform** GitHub
  - Built a full-stack AI-powered CRM platform to centralize multi-channel leads, manage customer pipelines, prioritize high-intent prospects, and provide AI-assisted sales intelligence.
  - Developed a React/TypeScript frontend and FastAPI/Python backend with PostgreSQL/Supabase, implementing lead capture, pipeline management, AI priority workflows, follow-ups, natural-language search, analytics, authentication, and customer/product management.
  - Integrated Google Gemini and LangGraph for AI-assisted sales analysis and designed a production-oriented architecture using SQLAlchemy, Alembic, Redis/Celery, Docker, automated testing, and role-based application structure.
- **SkillMap AI Agent with MCP — Agentic Career Intelligence** GitHub
  - Developed an AI-powered career intelligence agent using Google Gemini, LangChain, and LangGraph to research real-time skill demand and provide context-aware career and job recommendations.
  - Integrated Tavily Search and JSearch API to combine live industry insights and job listings from multiple external data sources for career analysis.
  - Implemented Model Context Protocol (MCP) with Composio and dynamic tool discovery using `MultiServerMCPClient`; added LangGraph `MemorySaver`, asynchronous tool execution, and automated validation with Pytest.

RELEVANT COURSEWORK

- Machine Learning, Deep Learning, Artificial Intelligence, Data Structures and Algorithms, Database Management Systems, Operating Systems.

LANGUAGES

- English, Telugu, Hindi